Secure Peer-to-Peer Payment System on Android-Based Mobile Devices via Vibration and Acceleration (CS4514 - FYP Presentation)

IP Aaron (SID: 57847690)
Supervisor: Prof HANCKE, Gerhard Petrus

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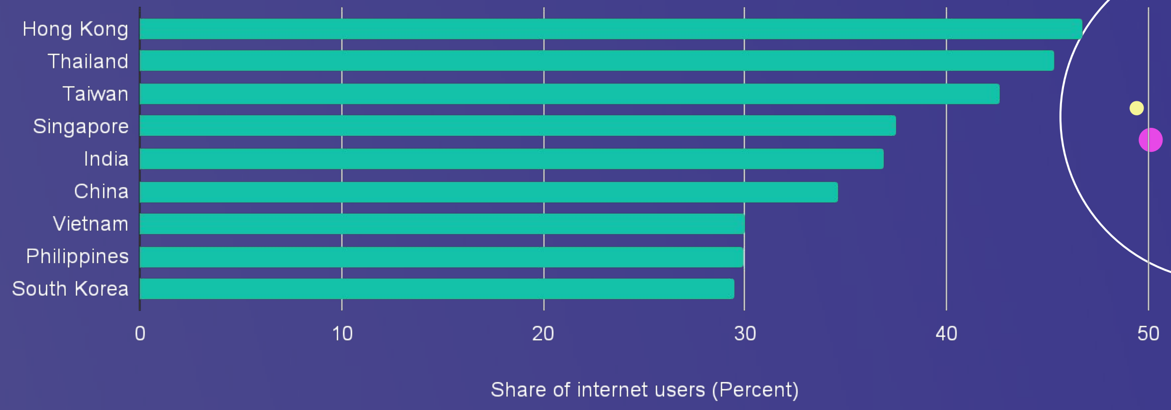
Introduction & Background

Background

P2P Mobile Payment System has become keystone of modern digital payment transactions

- Recent COVID-19 pandemic has increased the acceptance of mobile payments
- P2P payment market valued at approximately $9000 billion by 2030 globally
- Hong Kong has the highest percentage (46.7%) of interested users who use mobile payment system services monthly in the Asia-Pacific



Problem Statement

Near Field Communication (NFC)

- An Radio-frequency (RF) technology that supports short range wireless payments
- Commonly used in the P2P mobile payment system
- Such as Octopus and Apple Pay

Problem of NFC?

- Requires particular hardware
- Vulnerable to compromise such as relay attacks
- Drawbacks have encouraged the use of different channels for P2P payment!

Project Objectives

Overcome the challenges of NFC

Purpose an offline physical P2P mobile payment system without radio

Vibration communication channel

Examined the feasibility of the vibration-based P2P mobile payment system

Security protocols

Design a simple Lightweight Cryptographic Framework to fulfill the security goals of mobile payment system

Literature Review

Existing P2P Payment Solution (Except for Vibration)	Eavesdropping	Man-in-the-Middle (MITM)	Denial of Service (DOS)
NFC	High (RF signals can be intercepted)	High (Cloud-based NFC)	Moderate (RF jamming possible)
Bluetooth	High (Bluetooth sniffer can intercept the raw data)	High (MAC spoofing, pairing vulnerabilities)	High (RF noise can block paring)
Vibration-Based Channel	Low (Radio free, requires physical contact)	Low (Hard to intercept physical vibrations)	Low (No RF interference)

(Fahrianto et al., 2016; Hwang et al., 2012; Minar & Tarique, 2012; Maatallaoui et al., 2023)

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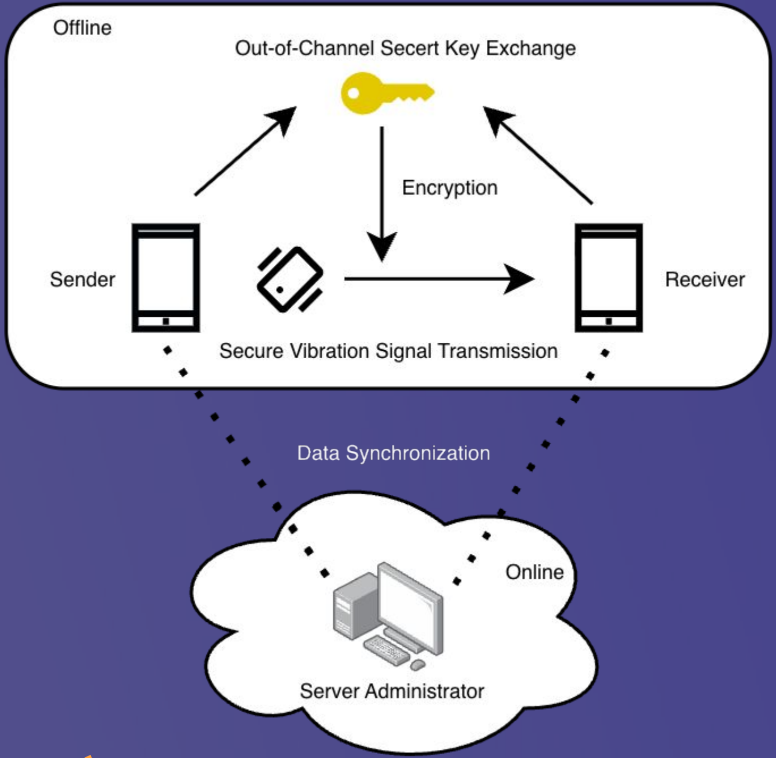
System Design & Methodology

Proposed Solution

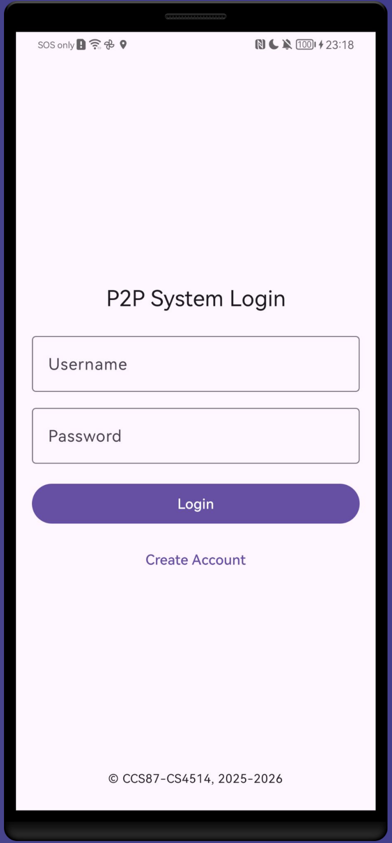


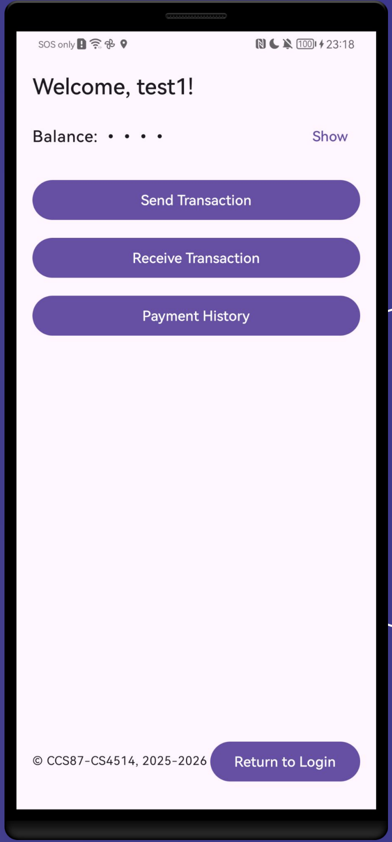
Proof-of-concept application

- Android Studio IDE / Kotlin
- 3 Major Components
- Functionalities of the mobile payment system (UI)
- Vibration Communication channel for transaction
- Security protocols in communication channel

System Design (UI)

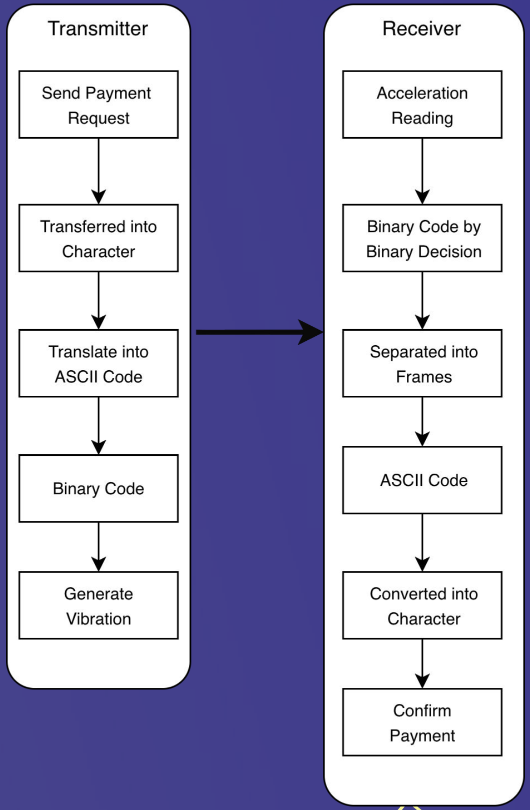
- Toolkit: Jetpack Compose
- Login/logout function
- Registration
- Send/ Receive payment using vibration
- Payment History
- Evaluation: System functionality test





System Design (Vibration)

- API: Android Haptic APIs & Android Sensor Framework
- Used Built-in Vibrator and Accelerometer
- On-Off Keying (OOK)-based protocol for encoding
- Adaptive thresholding Algorithm and simple binary for decoding
- Evaluation: Performance Testing (Accuracy)
- Novel P2P Medium: Enables P2P transaction via physical vibration patterns—a largely unexplored approach



Vibration Encode/Decode algorithm

Algorithm 1 Vibration encoding algorithm

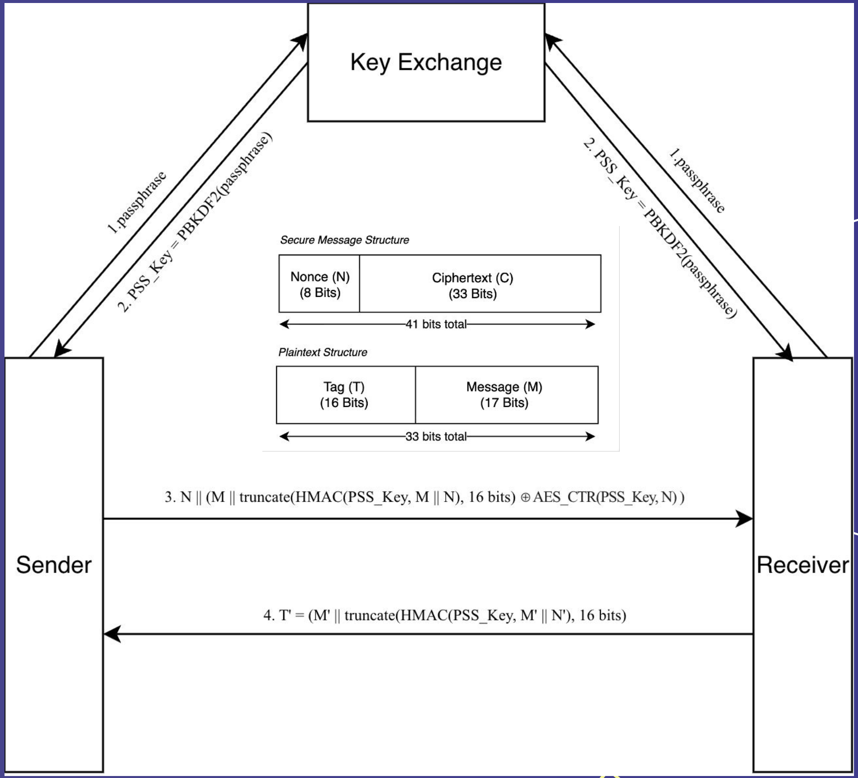
```
Require: Command character C
Ensure: Data Frame D
asciiCode ← ASCII(C)
binaryData ← ToBinary(asciiCode, length = 7)
startMarker ← "00010"
endMarker ← "00011"
fullBinary ← startMarker ‖ binaryString ‖ endMarker
D ← ∅
for all bit in fullBinary do
    if bit == '1' then
        Append 400ms to D
        Append 200ms to D
        Append 400ms to D
    else
        Append 1000ms to D
    end if
end for
return D
```

Algorithm 2 Vibration Decoding Algorithm

```
Require: List of Accelerometer datas A, Threshold τ = 9.65
Ensure: Decoded Commands C
Stage 1: Bit Reconstruction
Bits ← ∅
T_bit ← 1000ms
for each time window W of duration T_bit in A do
    count ← Number of samples in W where value > τ
    maxVal ← Maximum sample value in W
    if count > 5 AND maxVal > τ then
        Append '1' to Bits
    else
        Append '0' to Bits
    end if
    if Length of Bits ≥ 17 then
        break
    end if
end for
Stage 2: Frame Matching
StartMarker ← "00010"
EndMarker ← "00011"
C ← ∅
for i ← 0 to Length(Bits) do
    window ← Bits[i ... i + 17]
    if window starts with StartMarker AND ends with EndMarker then
        DataFrame ← window[5 ... 11]
        AsciiVal ← BinaryToDecimal(DataFrame)
        Command ← Char(AsciiVal)
        if Command ∈ {'a','b'} then
            Add Command to C
        end if
    end if
end for
return C
```

System Design (Security)

- API: javax.crypto/security
- Out-of-bound Secret Key Establishment
- Confidentiality: AES-CTR
- Integrity: HMAC tag & Nonce
- Evaluation: Penetration (Security) Testing
- Custom & Hybrid LWC Framework: Designed specifically to secure low-speed vibration channels (addressing a major research gap)



Typical operation performed behind LWC

```
Key derivation requested
Passphrase length=1
Salt (static) length=16 bytes
Derived PSS key length=16 bytes
Derived PSS key (hex)=ef411e1f82275d96f9c750af9b64bca8
=== Secure Sender Build Start ===
M (17-bit)=00010110000100011
Nonce N (0..255)=224
N (8-bit)=11100000
T (16-bit, truncated HMAC-SHA256)=0011111101110011
P = M || T (33-bit)=000101100001000110011111101110011
K (33-bit keystream via AES-CTR)=000100010110011000011100100011001
C = P XOR K (33-bit)=000001110111011110000011001101010
Payload = N || C (41-bit)=11100000000001110111011110000011001101010
=== Secure Sender Build End ===
```

```
[34071] COMPLETED BIT DETECTION: 41 bits
[34071] RAW BINARY DETECTED: 11100000000001110111011110000011001101010 (41 bits)
rawBits(len=41)=11100000000001110111011110000011001101010
[34072] SECURE RX: payloadBits41=11100000000001110111011110000011001101010
[34072] SECURE RX: N(bits)=11100000 N(int)=224
[34072] SECURE RX: C(33)=000001110111011110000011001101010
secureRx payload41=11100000000001110111011110000011001101010
secureRx nonceBits8=11100000 nonce=224
secureRx ciphertext33=000001110111011110000011001101010
[34075] SECURE RX: K(33)=000100010110011000011100100011001
[34075] SECURE RX: P'(33)=C XOR K =000101100001000110011111101110011
secureRx keystream33=000100010110011000011100100011001
secureRx plaintext33=000101100001000110011111101110011
[34075] SECURE RX: M'(17)=00010110000100011
[34075] SECURE RX: T_recv(16)=0011111101110011
[34076] SECURE RX: T_exp(16)=0011111101110011
secureRx tagRecv=0011111101110011 tagExp=0011111101110011
[34076] SECURE RX: TAG OK → decode legacy 17-bit message
[34076] FINAL DECODE: Searching for pattern in 17 bits
[34076] FOUND COMMAND: 'a' at index=0 (ASCII=97)
[34076] SUCCESS: Single command 'a' decoded
```

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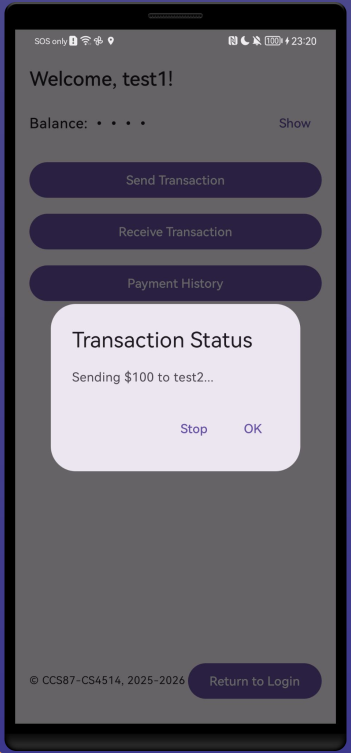
Evaluation and Results

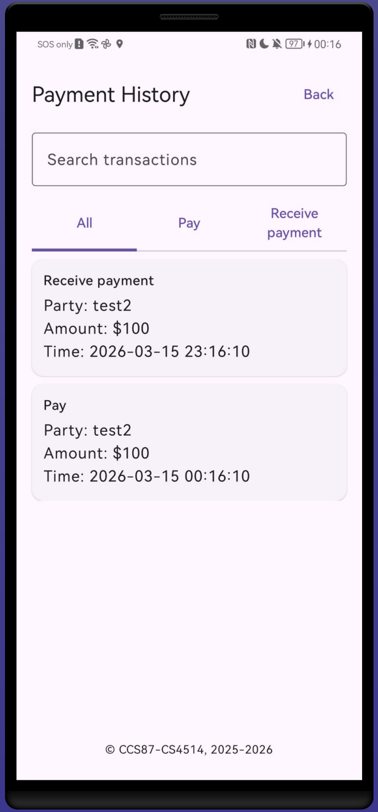
System functionality test (UI & Interaction)

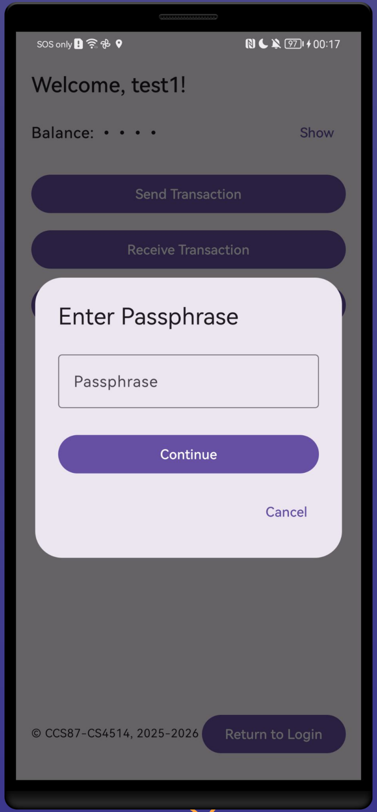
All user test cases passed

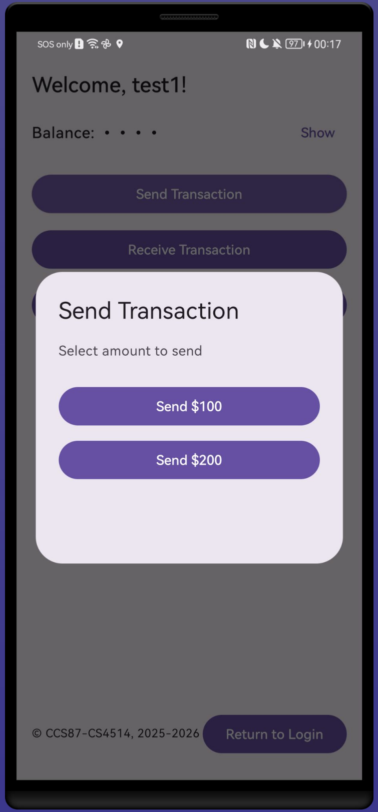
- User can perform the application as a financial application
- Sender can encode and encrypt the transaction request into a vibration pattern
- Receiver recognizes and decrypts it to the payment command
- The proposed vibration and LWC protocols successfully enable secure P2P payments



System functionality test (Cont)







Performance Testing (Magnitude-Based Threshold)

Optimal Decoding Threshold: 9.65 – 9.75

- **High & Stable Accuracy:** Transactions succeed reliably whether LWC is used or not

Threshold > 9.75 (Transaction Failure)

- **Without LWC (~65%):** Static, predictable patterns fail the same way every time
- **With LWC (Unstable):** Cryptographic randomness (nonce) creates complex patterns the receiver cannot decode

Threshold	Without LWC Framework	With LWC Framework
9.65	>95%	>95%
9.75	>95%	>95%
9.8	~77%	~54%
9.9	~65%	~49%
10	~65%	~36%

Performance Testing (Medium Objects)

High Accuracy / Success (Wood & Metal)

- Solid, high-density materials
- Low acoustic damping preserves the vibration signal

Low Accuracy / Fails (Cushion & Cotton)

- Soft, porous materials
- Absorbs and weakens high-frequency vibrations, preventing accurate decoding

Materials	Without LWC Framework	With LWC Framework
Wood Table	>95%	>95%
Metal Shelf	>95%	>95%
Cushioned chair	~85%	~68%
Cotton bed	~65%	~51%

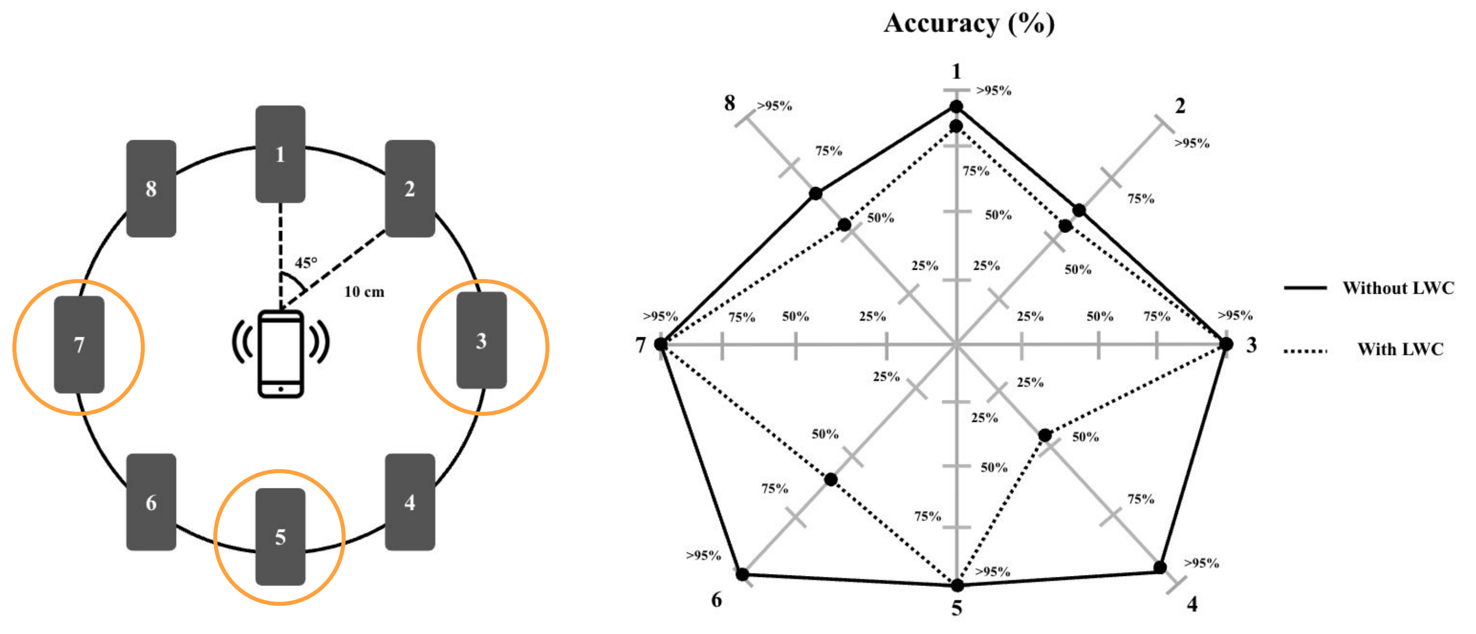
Performance Testing (Communication Distance)

Effective Communication Distance

- **Optimal Range (<= 20cm):** Robust, high-accuracy vibration transfer
- **Failure Range (> 30cm):** Severe signal loss; transactions reliably fail to decode

Distance	Without LWC Framework	With LWC Framework
10cm	>95%	>95%
20cm	>95%	>95%
30cm	~65%	~41%
40cm	~65%	~32%

Performance Testing (Communication Directions)



(a) Overall Settings

(b) Performance Results

- Internal hardware placement and device casing cause smartphones to emit vibration energy non-uniformly across different directions
- The successful transactions require strict physical alignment between the sender's vibration motor and the receiver's accelerometer

Penetration Testing & Security Validation

Key Management

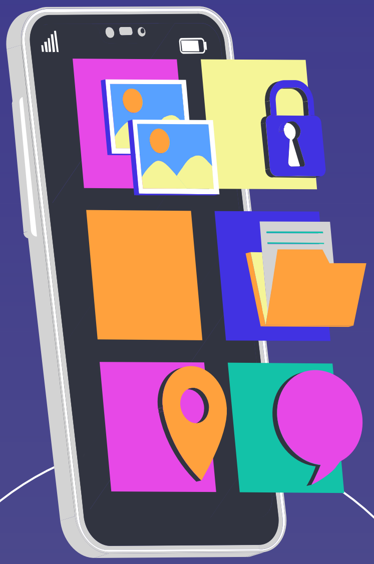
- Zero Key Reuse: A *ClearKey* function automatically wipes the PSS key after transaction completion, timeout, or cancellation

Threat Mitigation

- Eavesdropping Prevented (Confidentiality): AES-CTR encryption keeps the payload confidential. Additionally, the strict <= 20cm physical range ensures attackers would be physically visible to users
- Replay Attacks Blocked (Integrity): A unique 8-bit nonce and 16-bit HMAC tag ensure that recorded, replayed vibration patterns are immediately rejected

Other Security Goals Achieved

- Authentication & Non-repudiation: The shared passphrase (PSS Key) and unique nonce guarantee data origin and uniquely link the user to the transaction



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Reflection and Future Work

Limitation

Performance Trade-off (LWC Overhead)

- The LWC framework decreases overall stability and performance because the cryptographic payload (HMAC, nonce) increases the data size and randomness

Availability Vulnerability (Physical DoS)

- An attacker in proximity can inject random vibration patterns into the surface to intentionally jam the signal and block transactions

Further Work

Hardware Integration

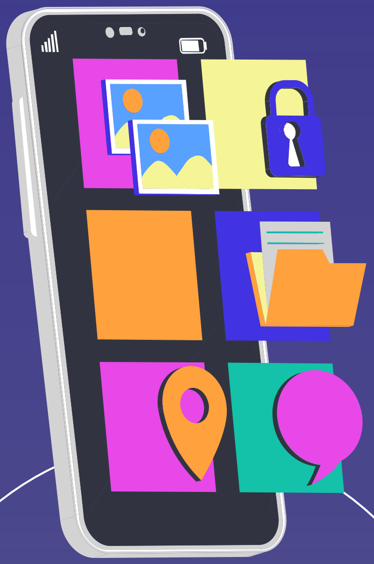
Integrate microphones and gyroscopes via sensor fusion to capture high-frequency signals and reduce environmental noise

Algorithm Enhancement

Upgrade the decoding algorithm with adaptive amplification and frequency adjustments to maximize the LWC framework's accuracy

Interference Mitigation

Implement a collision-detection mechanism to identify and block physical signal interference from unknown sources



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Conclusion

Conclusion

Problem

Increased usage of P2P mobile payment System but NFC is not secure…

Solution

We proposed a radio free approach, vibration-based transaction, the project tried to evaluate the feasibility of using it in practical secure financial application

Results

Our proposed secure P2P payment system using the vibration channel is reliable under certain circumstances and secured on prevent attacks and fulfill security goals

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Thanks!

Do you have any questions?